

Unit 5.5 Indefinite Integrals

Note Title

2012/05/16

Stewart Section 5.4 p297-400

Definition of an indefinite integral

$\int f(x) dx$ is called an indefinite integral and

$\int f(x) dx = F(x) + C$
if and only if $F'(x) = f(x)$.

Properties of indefinite integrals are in your text book and in your study guide.

- $\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$
- $\int [f(x) - g(x)] dx = \int f(x) dx - \int g(x) dx$
- $\int c f(x) dx = c \int f(x) dx$

Integral of a constant

- $\int c dx = cx + C$

Integral of a power function

$$\bullet \int x^n dx = \frac{1}{n+1} x^{n+1} + \underline{C} \quad (n \neq -1)$$

$$\bullet \int [f(x)]^n \cdot f'(x) dx = \frac{1}{n+1} [f(x)]^{n+1} + C$$

$(n \neq -1)$

Integral of the natural exponential function

$$\bullet \int e^x dx = e^x + C \quad \bullet \int e^{f(x)} \cdot f'(x) dx = e^{f(x)} + C$$

Integral of any exponential function

$$\bullet \int a^x dx = \frac{1}{\ln a} a^x + C \quad \bullet \int a^{f(x)} \cdot f'(x) dx = \frac{1}{\ln a} a^{f(x)} + C$$

The integrals of the trigonometric functions

$$\begin{aligned} \bullet \int \sin x dx &= -\cos x + C & \bullet \int \sin(f(x)) \cdot f'(x) dx &= -\cos f(x) + C \\ \bullet \int \cos x dx &= \sin x + C & \bullet \int \cos f(x) \cdot f'(x) dx &= \sin f(x) + C \\ \bullet \int \sec^2 x dx &= \tan x + C & \bullet \int \sec^2 f(x) \cdot f'(x) dx &= \tan f(x) + C \\ \bullet \int \operatorname{cosec} x \cot x dx &= -\operatorname{cosec} x + C \end{aligned}$$

$$\bullet \int \operatorname{cosec} f(x) \cdot \cot f(x) \cdot f'(x) dx = -\operatorname{cosec} f(x) + C$$

$$\bullet \int \sec x \tan x dx = \sec x + C$$

$$\bullet \int \sec f(x) \tan f(x) \cdot f'(x) dx = \sec f(x) + C$$

$$\bullet \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\bullet \int \operatorname{cosec}^2 f(x) \cdot f'(x) dx = -\cot f(x) + C$$

Integrals of the inverse trigonometric function

$$\bullet \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C = \sin^{-1} x + C$$

$$\bullet \int \frac{f'(x)}{\sqrt{1-(f(x))^2}} dx = \arcsin(f(x)) + C$$

$$\bullet \int \frac{1}{1+x^2} dx = \arctan x + C = \tan^{-1} x + C$$

$$\bullet \int \frac{f'(x)}{1+[f(x)]^2} dx = \arctan f(x) + C$$

Integrals of logarithmic functions

- $\int \frac{1}{x} dx = \ln |x| + C$

- $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$

Integrals of hyperbolic functions
(self study)

Examples We can ask three types of questions.

1] Find the area between the graph of $f(x) = \sin x$, the x -axis and the lines

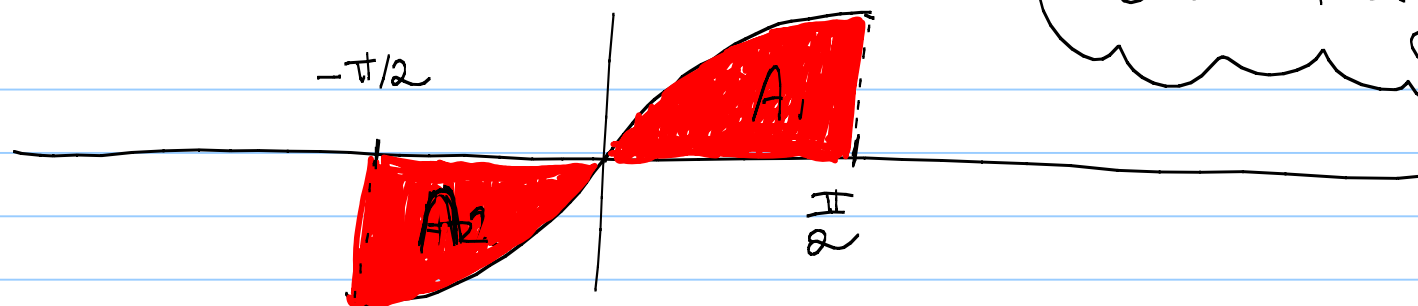
$$x = -\frac{\pi}{2} \quad \text{and} \quad x = \frac{\pi}{2}$$

2] Calculate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \, dx$

3] Find $\int \sin x \, dx$

Solutions

1]



You have to sketch a graph

$$\int_{-\frac{\pi}{2}}^0 \sin x \, dx = -\cos x \Big|_{-\frac{\pi}{2}}^0$$

$$= -\cos 0 - (-\cos(-\frac{\pi}{2}))$$

$$= -1 + 0$$

$$= -1$$

$$\int_0^{\frac{\pi}{2}} \sin x \, dx = -\cos x \Big|_0^{\frac{\pi}{2}}$$

$$= -\cos \frac{\pi}{2} - (-\cos(0))$$

$$= -0 + 1$$

$$= 1$$

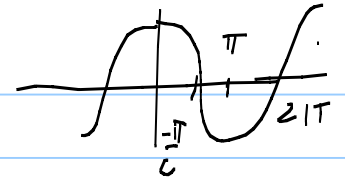
$$\text{Area} = 2$$

1

$$2) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \, dx = -\cos x \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= -\cos\left(\frac{\pi}{2}\right) - \left(-\cos\left(-\frac{\pi}{2}\right)\right)$$

$$= -0 + 0 = 0$$



$$3) \int \sin x \, dx = -\cos x + C \rightarrow$$

NB We do not use the method of substitution as given in textbook.

You must integrate either by using differentiation or using table of standard integrals.

Examples

Find

1. $\int x^5 dx$

2. $\int (1-2x)^5 dx$

3. $\int \sin^5 2x \cos 2x dx$

4. $\int e^{3t} dt$

5. $\int x^3 e^{x^4} dx$

6. $\int e^{\tan 3x} \sec^2 3x dx$

7. $\int e^{3x} \sec^2 e^{3x} dx$

8. $\int \frac{1}{1-2x} dx$

9. $\int \frac{x}{1+x^2} dx$

10. $\int \frac{3}{1+x^2} dx$

11. $\int \frac{x}{9+x^4} dx$

12. $\int \frac{1}{\sqrt{1-7x}} dx$

$$13. \int \frac{1}{\sqrt{1-4x^2}} dx$$

$$14. \int \frac{x}{\sqrt{1-4x^2}} dx$$

$$15. \int \frac{3^x}{\sqrt{1-3^x}} dx$$

$$16. \int \frac{e^{3x}}{\sqrt{1-e^{6x}}} dx$$

$$17. \int \frac{1}{(1-2x)^2} dx$$

$$18. \int \frac{\sin 3x}{1-\cos 3x} dx,$$

$$19. \int \frac{\sin 3x}{\sqrt{1-\cos 3x}} dx$$

Solutions:

$$1. \frac{1}{6} x^6 + C$$

$$2. = \frac{1}{6} (1-2x)^6 \cdot \left(-\frac{1}{2}\right) + C$$
$$= -\frac{1}{12} (1-2x)^6 + C$$

$$3. \int (\sin 2x)^5 \cdot \cos 2x \, dx$$
$$= \frac{1}{6} (\sin 2x)^6 \cdot \frac{1}{2} + C$$

$$4. \frac{1}{3} e^{3x} + C$$

$$5. \frac{1}{4} e^{x^4} + C$$

$$6. e^{\tan 3x} \cdot \frac{1}{3} + C$$

$$7. \tan e^{3x} \cdot \frac{1}{3} + C$$

$$8. \ln|1-2xc| \cdot \left(-\frac{1}{2}\right) + C$$

$$9. \frac{1}{2} \ln(x^2+1) + C$$

$$10. 3 \arctan x + C$$

$$11. \int \frac{x}{9+x^4} dx = \int \frac{x}{9(1+(\frac{x^2}{3})^2)} dx$$

$$= \frac{1}{9} \arctan\left(\frac{x^2}{3}\right) \cdot \frac{3}{2} + C$$

$$12. \int \frac{1}{\sqrt{1-7x}} dx = -\frac{1}{7} \cdot 2 (1-7x)^{-\frac{1}{2}} + C$$

$$13. \int \frac{1}{\sqrt{1-(2xc)^2}} dx = \frac{1}{2} \arcsin(2xc) + C$$

$$14. \int \frac{x}{\sqrt{1-4x^2}} dx = \int x (1-4x^2)^{-\frac{1}{2}} dx$$

$$= -\frac{1}{8} (1-4x^2)^{\frac{1}{2}} + C$$

$$15. \int \frac{3^x}{\sqrt{1-3^x}} dx.$$

$$= \int 3^x (1-3^x)^{-\frac{1}{2}} dx$$

$$= \frac{1}{\ln 3} (1-3^x)^{\frac{1}{2}} + C.$$

$$16. \int \frac{e^{3x}}{\sqrt{1-(e^{3x})^2}} = \frac{1}{3} \arcsin(e^{3x}) + C$$

$$17. \int \frac{1}{(1-2x)^2} dx = -\frac{1}{4x-2} + C$$

$$18. \int \frac{\sin 3x}{1-\cos 3x} dx = \frac{1}{3} \ln |\cos 3x - 1| + C$$

$$19. \int \frac{\sin 3x}{\sqrt{1-\cos 3x}} dx = \frac{2}{3} \sqrt{1-\cos 3x} + C$$

Examples Find

$$1. \int e^{5x} + x e^{x^2} dx$$

$$2. \int \tan 2x dx$$

$$3. \int \frac{\sec^2 \sqrt{x}}{\sqrt{4 + \tan \sqrt{x}}} dx$$

$$4. \int \frac{1}{4 + x^2} dx$$

$$5. \int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

$$6. \int \frac{\sin 6x}{(1 + \cos 6x)^3} dx$$

$$7. \int 8^{\operatorname{cosec} 5x} \operatorname{cosec} 5x \cot 5x \, dx$$

$$8. \int \frac{\sin 4x}{1 + \cos^2 4x} \, dx$$

$$9. \int \frac{x^2}{\sqrt{16 - x^6}} \, dx$$

